

GIHEKE TSS
END TERM I EXAMEN, QUESTION PAPER,
ACADEMIC YEAR:2024/2025
SECTOR: TECHNICAL SERVICES
TRADE: L4 ETE
MODULE TITLE: NETWORK FUNDAMENTALS
MAX: 100 Marks
DATE: 20/12/2024
DURATION:9h00-12h00

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of **Three ,Sections (A, B,C)**
 Follow the instructions given below, and answer the indicated

Section A:

Section B:

Section C:

SECTION A: Attempt All questions(55marks)

01	What is OSI model?	2marks																		
02	Arrange all layers of OSI models from first to last	3.5marks																		
03	Which layer of OSI model that offer Ip addressing?	0.5marks																		
04	Compare TCP and UDP with the following parameters <table border="1"> <thead> <tr> <th>Features</th><th>TCP</th><th>UDP</th></tr> </thead> <tbody> <tr> <td>Connection types</td><td></td><td></td></tr> <tr> <td>Reliability</td><td></td><td></td></tr> <tr> <td>Ordering</td><td></td><td></td></tr> <tr> <td>Flow control</td><td></td><td></td></tr> <tr> <td>speed</td><td></td><td></td></tr> </tbody> </table>	Features	TCP	UDP	Connection types			Reliability			Ordering			Flow control			speed			5marks
Features	TCP	UDP																		
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05	Differentiate network media types?	4marks																		
06	How can you differ broadband and baseband transmission technologies	2marks																		
07	Differentiate microwaves and infrared as types of wireless transmission techniques	3marks																		
08	What is subnet mask?	3mark																		
09	Write default subnet mask on class B and C Ip address	2marks																		
10	Write in full i. FLSM ii. VLMS	3mark																		
11	Differentiate types of Ip address with example ?	6mark																		
12	Define network protocol, after defining it provide four (4) examples	5marks																		
13	As network designer you are hired for implementing mesh topology which composed by 15 nodes so how can respond the following questions	4marks																		

	I. number of all port in the network II. b)number of port for each node? III. number of cables for entire network?	
14	What is prefix in subnetting ?	3marks
15	What is block size?	3marks
16	Write in full form a) VLSM b) FLSM	3marks
17	what is importance of Nat in network?	3marks

Section B: Attempt any three (3) questions (30 marks)

18	What is the number of host on the following subnet mask? a. /24 b. /25 c. /26 d. 27 e. /28	(10 marks)																
19	As network administrator you have this network id 192.168.2.0 with default subnet mask of 255.255.255.0 and you have been asked to subnet it into eight(8) subnetwork, so what will be the block size, the new subnet mask ,number of hosts for each subnetwork, and after first usable Ip address, last usable Ip address and broadcast Ip address for fourth subnetwork?	(10 marks)																
20	Observe the following table and respond the question accordingly <table><tr><td>Ip address Classes</td><td>Default subnet mask</td><td>Number networks</td><td>Number of hosts</td></tr><tr><td>Class A</td><td>?</td><td>?</td><td>?</td></tr><tr><td>Class B</td><td>?</td><td>?</td><td>?</td></tr><tr><td>Class C</td><td>?</td><td>?</td><td>?</td></tr></table>	Ip address Classes	Default subnet mask	Number networks	Number of hosts	Class A	?	?	?	Class B	?	?	?	Class C	?	?	?	(10 marks)
Ip address Classes	Default subnet mask	Number networks	Number of hosts															
Class A	?	?	?															
Class B	?	?	?															
Class C	?	?	?															

Section C: Attempt one (1) questions (15 marks)

21	Differentiate the following wireless transmission techniques and specify real life application of them? I. Infrared II. Microwaves III. Radio waves IV. Bluetooth V. Wi-fi	(15 marks)
22	Provide Tools, materials and equipment in VoIP system installation?	(15 marks)

END OF ASSESSMENT